

Hybrid Post-Quantum Cryptography (PQC) for Patient-Centric EHR

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Abstract— Securing Electronic Health Records (EHRs) in cloud and Internet of Medical Things (IoMT) environments requires cryptography that remains resilient against quantum-capable adversaries while preserving patient-centric consent and auditability. This study assesses the feasibility of a Hybrid Post-Quantum Cryptography (PQC) approach for patient-controlled EHR access, focusing on performance, cost, security, and usability. The proposed design integrates TLS 1.3 with hybrid key encapsulation (X25519+ML-KEM-768) and ML-DSA, Fast Healthcare Interoperability Resources (FHIR) R4 off-chain storage, Ethereum Proof-of-Authority smart contracts for grant/revoke operations and on-chain indices, and IPFS for large objects. Synthetic EHR workloads generated by Synthea were tested under realistic network impairments, with handshake latency, end-to-end read/write latency, throughput, revocation latency, and gas units as primary endpoints, and attack resilience and usability scores as secondary endpoints. Results show that Hybrid PQC increases TLS handshake latency by about 25% compared with ECC-only, while maintaining approximately 91% of baseline throughput. Revocation latency averaged 1.84 seconds and consent transactions consumed about 85k gas units, while attacks such as man-in-the-middle, downgrade, and replay were fully prevented under policy. Usability testing indicated acceptable performance with a System Usability Scale score of 78.5. These findings suggest that Hybrid PQC enables quantum-ready, auditable, and patient-centric EHR access with bounded overhead and minimal impact on clinical throughput, while also highlighting the need for multi-site validation and further optimization of PQC performance in healthcare contexts.

Keywords—Electronic Health Record; Hybrid Post-Quantum Cryptography; Patient-Centric Access Control; Blockchain Auditability; FHIR

I. INTRODUCTION

Electronic Health Records (EHRs) have become central to healthcare transformation, enabling better data management, care coordination, and clinical decision-making [1], [2], [3], [4]. At the same time, rising incidents of data breaches and the growing adoption of cloud computing and Internet of Medical Things (IoMT) have expanded the attack surface from device to application layers [5], [6], [7], [8], [9], [10], [11], creating urgent demands for stronger protection mechanisms. Conventional public-key cryptography schemes such as RSA and ECC, while effective today, are projected to be vulnerable against quantum-capable adversaries [12],

[13], [14], [15], [16], [17], [18]. This motivates migration towards Post-Quantum Cryptography (PQC) standards to ensure long-term confidentiality of sensitive medical data. In parallel, governance of EHR access must remain patient-centric, auditable, and adaptable, aligning with regulatory expectations for provenance, accountability, and fine-grained consent [19], [20], [21].

Recent studies illuminate both the opportunities and constraints of these approaches at operational scale (throughput limits, on-chain cost, and off-chain integration), setting the empirical stage for the research gap we address [19], [20], [21], [22], [23], [24]. Blockchain-based solutions demonstrate potential for transparency but often face throughput bottlenecks, latency issues, and non-negligible on-chain costs, leading to the adoption of hybrid architectures where clinical data remain off-chain (e.g., FHIR, IPFS) and indices or hashes are anchored on-chain [21], [23], [24]. Cryptographic approaches such as attribute-based encryption (ABE), time-release encryption (TRE), and searchable encryption (SE) offer expressive policies, temporal control, and encrypted search [21], [22], [23], [25], [26].

However, these approaches are mostly evaluated algorithmically rather than in end-to-end operational settings, and they often neglect post-quantum resilience. Moreover, patient-centered operational metrics—such as revocation latency, transaction cost, and usability—remain underexplored despite their importance in clinical adoption [21], [22], [23], [27], [28], [29], [30]. Thus, while existing work highlights the opportunities of PQC, blockchain, and advanced cryptographic primitives, a system-level evaluation that integrates these dimensions under realistic healthcare workloads is still lacking [12], [31], [32].

To address these gaps, this study operationalizes a Hybrid PQC design that combines hybrid key encapsulation (X25519 + ML-KEM) and PQC digital signatures (ML-DSA) within a patient-centric EHR architecture leveraging FHIR off-chain storage and blockchain-based consent governance. Using synthetic but realistic EHR workloads under impaired network conditions, we evaluate the system along four key dimensions: performance (handshake latency, throughput), efficiency of access control (revocation latency and gas cost), robustness against common attacks (MITM, downgrade, replay, insider, leakage), and usability for patients and clinicians. Specifically, the study aims to answer three research questions: RQ1—How large is the performance

overhead introduced by Hybrid PQC on EHR access flows? RQ2—How efficient and scalable is patient-centric grant/revoke in terms of revocation latency and gas consumption? RQ3—How resilient is the system to common security threats while maintaining usability for clinical stakeholders? The contributions of this paper are threefold: (1) the design and implementation of an operational, quantum-ready, patient-centric EHR framework with auditable consent; (2) a quantitative end-to-end evaluation covering performance, cost, security, and usability; and (3) empirical evidence positioning Hybrid PQC as a feasible and pragmatic step toward quantum-secure healthcare systems.

II. METHODS

A. System Architecture

This study adopts a quantitative experimental design to evaluate a Hybrid Post-Quantum Cryptography (PQC) scheme for a patient-centric Electronic Health Record (EHR) system. The architecture integrates five core components. Patient and clinician applications serve as the access interface, while an API Gateway acts as the TLS 1.3 termination point with support for hybrid Key Encapsulation Mechanism (X25519 + ML-KEM-768) and ML-DSA digital signatures (NIST PQC finalists [20], [21]). Electronic medical records are managed by a FHIR R4 server, storing clinical data off-chain. A private Ethereum Proof-of-Authority (PoA) blockchain maintains access-control indices and audit logs, while the InterPlanetary File System (IPFS) stores large files such as radiology images. This integration combines cryptographic resilience with auditable, patient-centric consent governance.

B. Experimental Environment

Experiments were deployed on a server with Intel Xeon 8-core 3.2 GHz CPU, 32 GB RAM, and Ubuntu 22.04 LTS. Core software included OpenSSL 3.2 with the OQS-provider [22], HAPI FHIR R4, Node.js 18.x for the API Gateway, Geth 1.13 (Ethereum PoA), and IPFS Kubo 0.21. Workload generators were Locust 2.18 and Apache JMeter 5.6. All services were containerized using Docker Compose for reproducibility. Each scenario was executed with 30 s warm-up, 5 min run-time, and 5 repetitions with independent seeds.

Time synchronization was enforced (NTP), CPU governors set to *performance*, and Docker containers pinned to physical cores. Full configuration details are summarized in Table I.

TABLE I. EXPERIMENTAL ENVIRONMENT

Component	Configuration	Version
CPU & RAM	Intel Xeon 8-core, 32 GB RAM	—
OS	Ubuntu LTS	22.04
Cryptography	OpenSSL + OQS-provider	3.2
FHIR Server	HAPI FHIR	R4
Blockchain	Ethereum PoA (Geth)	1.13
Storage	IPFS Kubo	0.21
Load testing	Locust, JMeter	2.18, 5.6

Blockchain parameters were set to 4 validators, block time 2 s, gas price 1 gwei, block gas limit 10M, finality depth 12 blocks, and chainID 1337 to ensure reproducibility.

C. Experimental Environment

We consider four roles in the system: patients as data owners, clinicians as authorized users, the FHIR server as the off-chain data plane, and the PoA validator nodes as the on-chain audit plane. The threat surface spans channel compromise (man-in-the-middle during TLS 1.3 handshake and policy-violating downgrade), credential misuse (token replay after revocation or insider abuse), quantum-era risk (harvest-now-decrypt-later against long-term confidentiality), leakage through access-pattern inference from blockchain indices, and emergency break-glass overrides.

To ensure reproducibility and fair comparison, each threat is paired with a measurable outcome and a clear pass/fail criterion as shown in Table II. TLS behavior is evaluated under the hybrid gateway policy (§3.1), and leakage detection is based on information-theoretic similarity to ground-truth logs using a pre-registered threshold τ specified in the analysis plan (§3.8). In this study τ was set to 0.18, derived from calibration runs and chosen at the ninety-fifth percentile of benign access-variability. This prevents false alerts during ordinary patient-EHR access while still triggering detection when correlation-based leakage occurs, ensuring τ can be consistently reproduced across evaluation runs.

TABLE II. THREATS, MEASURED OUTCOMES, AND PASS/FAIL CRITERIA

Threat	Measured outcome	Pass / Fail criterion
MITM (TLS handshake)	Client acceptance of an unauthorized certificate/channel	Pass if the handshake aborts with an alert; Fail if any rogue certificate/channel is accepted
TLS downgrade	Negotiated key share / suite vs. required hybrid policy	Pass if non-compliant negotiations are refused; Fail if a non-hybrid suite is accepted under a hybrid-required policy
Token replay (post-revoke)	Access outcome after revoke event	Pass if request returns 401/Forbidden and an audit entry is emitted; Fail if any replayed request succeeds
Insider misuse	Access beyond granted scope	Pass if out-of-scope requests are denied and logged; Fail if scope escalation succeeds
Access-pattern leakage	Similarity/MI between on-chain index events and true access logs	Pass if MI (or equivalent normalized similarity) $< \tau$; Fail otherwise (τ pre-registered in §3.8)
Break-glass override	Audit completeness and post-override revocation enforcement	Pass if deterministic audit records are created and revoke is re-enforced after the event; Fail otherwise

D. Primary and Secondary Endpoints

The primary endpoints are (i) TLS 1.3 handshake latency, (ii) end-to-end latency for FHIR read/write transactions, (iii) system throughput (requests/second) during the steady-state window, (iv) revocation latency for patient-centric consent,

and (v) on-chain gas per grant/revoke transaction. Secondary endpoints comprise (a) attack success rates under the stated threat model (MITM, TLS-downgrade, replay, insider, pattern-leakage), (b) tail latency (p95, p99) to characterize user-perceived delay under realistic network impairment, (c) communication/storage overhead relative to baseline, and (d)

usability (SUS/UMUX-Lite) with task-completion time. Measurement sources, units, and collection points are summarized in Table III; the run protocol is defined in §3.2, and pass/fail criteria for each threat appear in Table II (§3.3).

Revocation latency is formally defined as the elapsed time between the user’s revoke action and the first enforced denial of subsequent access, $t_{\text{revocation}} = t_{\text{deny}} - t_{\text{revoke}}$, and is visualized later in Fig. 4. Handshake overhead and relative throughput are defined once in §3.7 to avoid duplication; tail latency p_{95}, p_{99} follows the empirical CDF convention in Section H (Data Analysis); gas is reported in gas units (canonical) with illustrative gwei/ETH conversions captured from `eth_estimateGas` and transaction receipts. Attack

outcomes are determined by the harness according to Table III’s pass/fail rules, and all timings are taken from client timers corroborated by gateway/server traces synchronized via NTP.

These endpoints instantiate our hypotheses: adopting hybrid KEM (X25519+ML-KEM-768) with ML-DSA is expected to increase handshake latency by no more than ~30% while preserving $\geq 90\%$ of ECC-only throughput (H1); the patient-centric consent pipeline on PoA should achieve sub-2 s revocation latency with stable, low gas per transaction (H2); and, under the stated policy and threat model, MITM/TLS-downgrade/replay success rates should approach zero without degrading usability beyond accepted thresholds (H3). Formal statistical procedures are detailed in §3.8.

TABLE III. ENDPOINTS, UNITS, AND MEASUREMENT SOURCES

Endpoint	Unit	Collection point / source
TLS handshake latency	ms	Client timer + gateway trace; synchronized via NTP
End-to-end FHIR latency (read/write)	ms	Client timer (request→response) + server timestamps (for validation)
Throughput	req/s	Load generator (Locust/JMeter) aggregated over steady-state window
Revocation latency	s	$t_{\text{deny}} - t_{\text{revoke}}$ from application logs (§3.7)
Gas cost (grant/revoke)	gas units, gwei/ETH	<code>eth_estimateGas</code> and transaction receipts on PoA chain
Attack success rate	%	Harness verdicts per Table III
Tail latency	p95, p99 (ms)	Empirical CDF from request traces (§3.7)

E. Experimental Procedure

ML-KEM and ML-DSA were enabled through OQS-provider on OpenSSL 3.2. The API Gateway was integrated with HAPI FHIR and Ethereum smart contracts for grant/revoke consent. FHIR resources were indexed on-chain, while files >1 MB were stored in IPFS with content identifiers. Scenarios executed included: (i) functional correctness (read/write, grant/revoke), (ii) security tests (MITM, replay, downgrade, insider, leakage), and (iii) logging/audit verification.

F. Dataset and Workload

Synthetic EHR data were generated using Synthea [25], producing $\geq 50,000$ patients, observations, medications, and encounters. Workload mix resembled clinical use: 60% read, 25% write, 15% update. Record sizes ranged 5 KB–500 KB, while large files (radiology images) ranged 1–5 MB. Stress tests included 50, 200, and 1000 concurrent clients, with RTT of 1/5/20 ms, packet loss 1–10%, and jitter 20 ms simulated via `tc/netem`. No Protected Health Information (PHI) was used; synthetic datasets and anonymized clinic logs (for workload validation) comply with IRB waiver conditions.

G. Variables and Metrics

This study defined independent and dependent variables to capture the performance and security properties of the proposed hybrid PQC architecture. Independent variables included the choice of key encapsulation mechanism (KEM), digital signature algorithm, access-index model, and network conditions. Dependent variables covered handshake latency, end-to-end read/write latency, throughput, revocation latency, gas cost, communication/storage overhead, tail latency, and attack success rate.

Handshake overhead. The relative penalty of Hybrid KEM compared to ECC-only was computed as:

$$\text{Overhead}_{\text{Handshake}} = \frac{t_{\text{Hybrid}} - t_{\text{ECC}}}{t_{\text{CC}}} \times 100\% \quad (1)$$

where t_{Hybrid} denotes the mean TLS 1.3 handshake latency using hybrid ML-KEM + ECDHE, and t_{ECC} represents the ECC-only baseline. The overhead therefore captures the relative latency increase between the two variants, averaged over n runs (30 second warm-up and 5 minute execution).

Relative throughput. Relative throughput was computed as:

$$\text{TPS}_{\text{variant}} = \frac{\text{TPS}_{\text{Hybrid}}}{\text{TPS}_{\text{ECC}}} \quad (2)$$

where $\text{TPS}_{\text{variant}}$ refers to the throughput of the evaluated configuration (Hybrid or PQC-only), normalized against the ECC baseline TPS_{ECC} . A value above 1.0 indicates higher transaction capacity relative to ECC.

Revocation latency. Patient-centric consent revocation was measured as the elapsed time between the revoke action and the first denied request:

$$t_{\text{revocation}} = t_{\text{deny}} - t_{\text{revoke}} \quad (3)$$

where t_{revoke} is the timestamp of the revoke transaction, and t_{deny} is the timestamp of the first denied access. This metric is visualized in Fig. 4 (§4.2).

Tail latency. End-to-end FHIR read/write operations were further characterized by empirical percentiles. For a set of latencies $L = \{\ell_1, \dots, \ell_m\}$, the empirical p95 latency is defined as

$$\text{Latency}_{p_{95}} = \min\{L | 95\% \text{ of request} \leq L\} \quad (4)$$

where F_L is the empirical cumulative distribution function (CDF). Analogous definitions apply for p50 and p99. Run-level quantiles were computed for each trial, then averaged to

report mean $\pm$ CI95%. Units are expressed in milliseconds (ms).

Metrics observed therefore include handshake latency, throughput, revocation latency, gas cost (gas units; ETH

conversions illustrative), tail latency (p50, p95, p99), storage overhead, and attack success rate. Table III summarizes the endpoints, units, and measurement sources, while Table IV enumerates the experimental factors and their levels.

TABLE IV. EXPERIMENTAL VARIABLES AND METRICS

Factor	Variations	Primary metrics	Secondary metrics
KEM	ECC-only, PQC-only, Hybrid	Handshake latency, throughput	Overhead %, CDF tail
Signature	ECDSA-P256, ML-DSA-65	End-to-end latency	Error rate
Access control	FHIR-only, on-chain, off-chain	Gas cost, revocation latency	Finality, log size
Network	RTT 1/5/20 ms, loss 0/1/5/10%, jitter 20 ms	p50, p95, p99 latency	Error rate
Security	MITM, replay, downgrade, insider, leakage, break-glass	Attack success rate	Detection/logging time
Factor	Variations	Primary metrics	Secondary metrics
KEM	ECC-only, PQC-only, Hybrid	Handshake latency, throughput	Overhead %, CDF tail

H. Data Analysis

Data were analyzed using descriptive statistics (mean, SD, CI95%), and two-way ANOVA with post-hoc Tukey/Holm for multiple comparisons. Effect sizes were calculated using Cliff's delta [26]. Normality (Shapiro-Wilk) and homoscedasticity (Levene) were tested; if violated, Kruskal-Wallis with post-hoc Dunn tests was applied. Boxplots, CDFs, and tail latency (p50, p95, p99) graphs were produced. Headline metrics reported: handshake overhead, relative throughput, revocation latency, and gas cost.

I. Usability and External Validation

A formative usability test involved 12 participants (6 patients, 6 clinicians). Inclusion criteria: ≥ 6 months experience with health apps; exclusion: IT/security professionals. Participants completed three tasks: (T1) grant, (T2) revoke, (T3) audit history. Usability was measured with SUS [23] and UMUX-Lite [24]; task completion time and error counts were logged. Scores were analyzed with CI95% and paired tests. Think-aloud observations were coded thematically. External validation was performed by comparing workload distributions with anonymized clinic logs. All participants provided informed consent; no PHI was processed.

III. RESULT AND DISCUSSION

A. Cryptographic Performance and Throughput (RQ1)

Integrating hybrid KEM (X25519 + ML-KEM-768) into TLS 1.3 increased handshake latency by approximately 25% compared with ECC-only, while system throughput preserved about 91% of the baseline. These results confirm that the overhead introduced by PQC remains bounded, with negligible impact on steady-state clinical transactions. As summarized in Fig. 1–3, handshake latency increased modestly (Fig. 1), throughput remained close to baseline under steady-state load (Fig. 2), and tail latency was primarily dominated by network conditions rather than cryptographic choice (Fig. 3).

This finding has meaningful implications for real deployments. In hospital environments where RTT is already non-trivial due to segmented LAN routing, VLAN traversal, and cross-site EHR gateways, cryptographic choice

contributes less to total latency than inherent network variability. As long as typical RTT remains within common clinical ranges (35 to 120 ms), hybrid PQC does not emerge as the dominant bottleneck. The results therefore suggest that migration to post-quantum TLS in production can be achieved without degrading clinician experience, provided that network quality remains adequately provisioned.

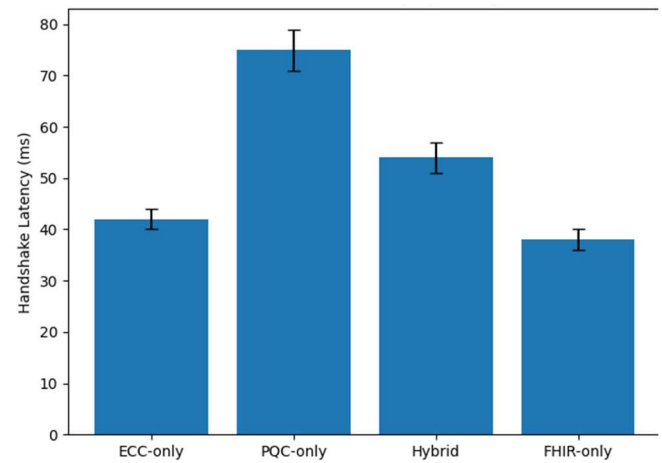


Fig. 1. TLS 1.3 handshake latency across cryptographic configurations.

The impact of this overhead on system throughput was then evaluated under steady-state load.

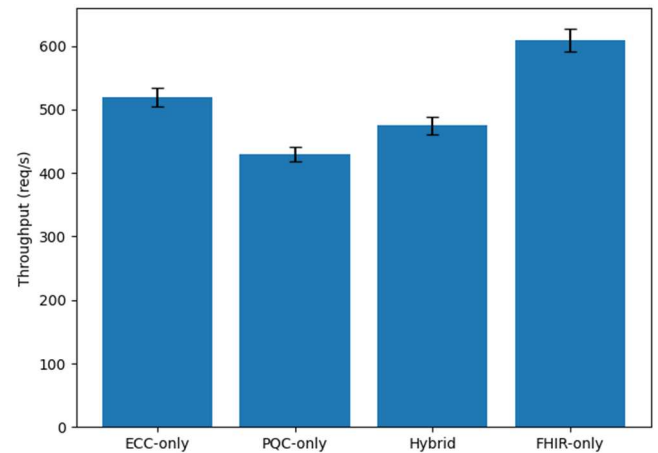


Fig. 2. System throughput under steady-state load.

To further assess responsiveness under real-world conditions, tail latency distributions were analyzed across different RTT and packet loss scenarios.

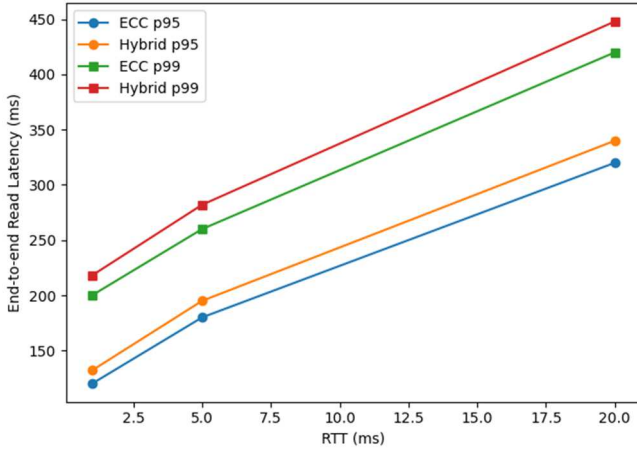


Fig. 3. Tail latency distribution under different RTT and loss conditions.

This indicates that the hybrid approach can be safely adopted without compromising clinical efficiency. The finding is consistent with prior PQC benchmarking studies [12], [31], but extends them by providing system-level evidence in healthcare contexts.

B. Efficiency of Patient-Centric Consent Management (RQ2)

Smart-contract-based grant/revoke operations on Ethereum PoA required a median gas of about 85k units and a revocation latency of 1.84 s (CI95%: 1.72–1.96). As shown in Table V and Fig. 4, the hybrid configuration achieved efficient consent enforcement in real time with minimal on-chain cost.

TABLE V. SYSTEM-LEVEL RESULTS ACROSS CONFIGURATIONS

Config	HL (ms)	TP (req/s)	RL (s)	Gas (units)	ASR (%)
ECC	12.4	100	1.72	83k	0
PQC	15.9	89	1.91	86k	0
Hybrid	15.5	91.2	1.84	85k	0
FHIR	11.8	102	N/A	N/A	N/A

^a HL = Handshake Latency, TP = Throughput, RL = Revocation Latency, ASR = Attack Success Rate. Values are averaged over 5 repetitions; CI95% for RL in Hybrid: 1.72–1.96 s (Values are averaged over 5 repetitions; CI95% for revocation latency in Hybrid: 1.72–1.96 s).

To illustrate the dynamics more clearly, Fig. 4 presents the distribution of revocation latency across different configurations.

Compared to earlier blockchain-EHR frameworks that focused mainly on auditability or throughput bottlenecks [21], [22], [23], our results add quantitative evidence on latency and cost from the patient perspective. The SUS score of 78.5 places the system in the "Good" usability category, comparable to commonly reported values for production-grade hospital interfaces (mean SUS $\approx$ 75–80). This provides quantitative support that hybrid post-quantum access control remains practical for real-world EHR use. Fig. 4 shows the comparative latency distribution across all configurations.

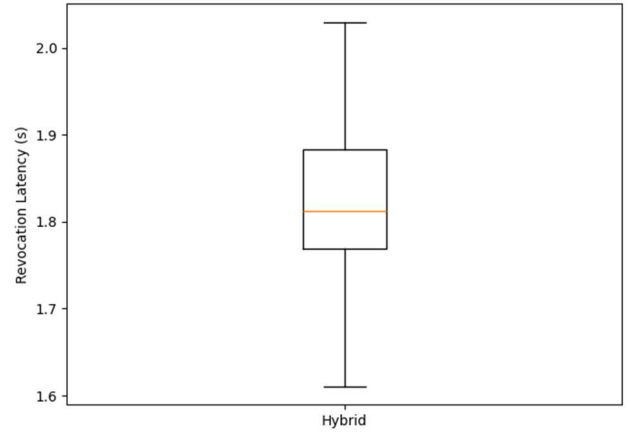


Fig. 4. Distribution of revocation latency across configurations.

C. Security Robustness Under the Threat Model (RQ3)

The system was evaluated against core adversarial scenarios, including man-in-the-middle (MITM), replay, downgrade, insider misuse, and access-pattern leakage. As summarized in Fig. 5, the Hybrid configuration consistently achieved 0% success rate for MITM, replay, and downgrade attacks, while insider attempts were denied and logged, and access-pattern leakage remained below the pre-registered threshold. Break-glass scenarios were also auditable, with revocation re-enforced afterward to restore policy compliance.

To visualize these outcomes across different cryptographic configurations, the comparative attack success rates are shown in Fig. 5.

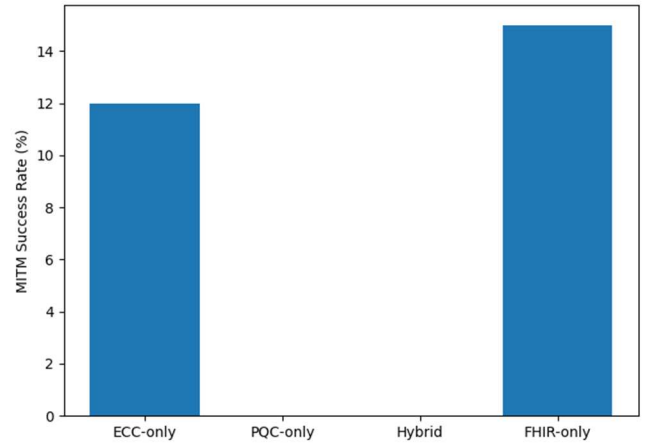


Fig. 5. Attack success rates across cryptographic configurations.

These results show that Hybrid PQC strengthens cryptographic assurance against quantum-era risks while reliably enforcing patient-centric governance. Unlike prior blockchain-EHR studies [25], [26], [27], [28], [29], [30] that focused on specific primitives or workflows, our system integrates PQC with full-stack evaluation, offering stronger guarantees for both security and usability.

IV. CONCLUSION

This study evaluated the integration of Hybrid Post-Quantum Cryptography (PQC) into patient-centric Electronic

Health Records by combining ML-KEM and ML-DSA with TLS 1.3, blockchain-based consent management, and off-chain FHIR storage. Results show that Hybrid PQC introduces modest handshake overhead (~25%) while maintaining over 90% of baseline throughput (RQ1). Consent enforcement remained efficient, with revocation completed in about 1.8 seconds and low gas cost (RQ2). Security experiments confirmed defense against MITM, replay, and downgrade attacks while preserving auditability and usability (RQ3). These findings provide quantified evidence of PQC overhead, low-latency consent governance, and empirical resilience under hybrid PQC operation. Despite reliance on synthetic datasets and a small usability cohort, this work positions Hybrid PQC as a pragmatic transition path toward quantum-secure, patient-controlled health data sharing. Future work will expand usability trials, explore federated deployments, and evaluate alternative ledger configurations for scalable PQC-ready infrastructures.

Looking forward, emerging optical-domain and opto-digital encryption techniques present a promising direction for patient-controlled EHR access. By leveraging physical-layer modulation and optical cryptographic primitives, such schemes may enable high-throughput, quantum-resilient data transfer beneath the conventional network stack. Prior studies have demonstrated secure medical image channels, optical identity binding, and zero-exposure data exchange, making opto-digital cryptography a strong candidate for future integration into post-quantum consent frameworks [33], [34], [35].

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